

EXPERIMENT 4

Aim:

Using the Employee schema, write queries with INNER JOIN, LEFT JOIN, self-join, 3-way join, correlated subqueries, EXISTS, and simulated INTERSECT and EXCEPT. Compare execution plans using EXPLAIN

Source Code:

```
DROP TABLE IF EXISTS Employee_Project;
```

```
DROP TABLE IF EXISTS Employee;
```

```
DROP TABLE IF EXISTS Project;
```

```
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (
```

```
    dept_id INT PRIMARY KEY,
```

```
    dept_name VARCHAR(50),
```

```
    location VARCHAR(50)
```

```
);
```

```
CREATE TABLE Project (
```

```
    project_id INT PRIMARY KEY,
```

```
    project_name VARCHAR(100),
```

```
    budget DECIMAL(12,2)
```

```
);
```

```
CREATE TABLE Employee (
```

```
    emp_id INT PRIMARY KEY,
```

```
    emp_name VARCHAR(50),
```

```
    job_title VARCHAR(50),
```

```
    salary DECIMAL(10,2),
```

```
    hire_date DATE,
```

```
    dept_id INT,
```

```
    manager_id INT,
```

```
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id),
```

```
    FOREIGN KEY (manager_id) REFERENCES Employee(emp_id)
```

```
);
```

```
CREATE TABLE Employee_Project (  
    emp_id INT,  
    project_id INT,  
    hours_worked INT,  
    PRIMARY KEY (emp_id, project_id),  
    FOREIGN KEY (emp_id) REFERENCES Employee(emp_id),  
    FOREIGN KEY (project_id) REFERENCES Project(project_id)  
);
```

```
INSERT INTO Department VALUES
```

```
(1, 'IT', 'Delhi'),  
(2, 'HR', 'Gurgaon'),  
(3, 'Finance', 'Mumbai'),  
(4, 'Marketing', 'Bangalore'),  
(5, 'Sales', 'Hyderabad');
```

```
INSERT INTO Project VALUES
```

```
(101, 'Cloud Migration', 500000),  
(102, 'E-Commerce Website', 350000),  
(103, 'Mobile Application', 250000),  
(104, 'HR Automation', 150000),  
(105, 'Financial Analytics', 450000),  
(106, 'Marketing Campaign', 200000),  
(107, 'CRM Development', 300000),  
(108, 'Cyber Security', 600000);
```

```
INSERT INTO Employee VALUES
```

```
(1, 'Amit', 'IT Manager', 90000, '2018-01-10', 1, NULL),  
(2, 'Neha', 'HR Manager', 85000, '2018-03-15', 2, NULL),  
(3, 'Rahul', 'Finance Manager', 95000, '2017-06-20', 3, NULL),  
(4, 'Priya', 'Marketing Manager', 88000, '2019-02-12', 4, NULL),  
(5, 'Arjun', 'Sales Manager', 92000, '2017-09-25', 5, NULL);
```

```
INSERT INTO Employee VALUES
```

```
(6, 'Ravi', 'Software Engineer', 60000, '2020-01-10', 1, 1),
```

(7, 'Simran', 'Software Engineer', 65000, '2021-04-15', 1, 1),
 (8, 'Karan', 'Database Developer', 70000, '2020-07-20', 1, 1),
 (9, 'Pooja', 'Cloud Engineer', 75000, '2022-01-12', 1, 1),
 (10, 'Vikas', 'System Engineer', 58000, '2021-09-25', 1, 1),
 (11, 'Anjali', 'HR Executive', 55000, '2020-02-10', 2, 2),
 (12, 'Rohit', 'Recruiter', 50000, '2021-05-15', 2, 2),
 (13, 'Kavya', 'HR Executive', 52000, '2022-06-20', 2, 2),
 (14, 'Manish', 'Training Officer', 60000, '2020-08-12', 2, 2),
 (15, 'Sneha', 'HR Analyst', 58000, '2021-10-25', 2, 2),
 (16, 'Deepak', 'Financial Analyst', 70000, '2019-02-10', 3, 3),
 (17, 'Nisha', 'Accountant', 62000, '2020-05-15', 3, 3),
 (18, 'Varun', 'Financial Analyst', 68000, '2021-06-20', 3, 3),
 (19, 'Meena', 'Accountant', 60000, '2022-08-12', 3, 3),
 (20, 'Sahil', 'Finance Executive', 57000, '2021-11-25', 3, 3),
 (21, 'Tanya', 'Marketing Executive', 55000, '2020-03-10', 4, 4),
 (22, 'Mohit', 'SEO Specialist', 60000, '2021-05-15', 4, 4),
 (23, 'Isha', 'Content Manager', 65000, '2022-07-20', 4, 4),
 (24, 'Akash', 'Marketing Analyst', 58000, '2020-09-12', 4, 4),
 (25, 'Riya', 'Social Media Manager', 62000, '2021-12-25', 4, 4),
 (26, 'Naveen', 'Sales Executive', 55000, '2020-04-10', 5, 5),
 (27, 'Komal', 'Sales Executive', 57000, '2021-06-15', 5, 5),
 (28, 'Yash', 'Sales Analyst', 60000, '2022-07-20', 5, 5),
 (29, 'Pankaj', 'Business Executive', 65000, '2020-10-12', 5, 5),
 (30, 'Divya', 'Sales Executive', 58000, '2022-01-25', 5, 5);

INSERT INTO Employee_Project VALUES

(1,101,100),
 (6,101,120),
 (7,102,100),
 (8,101,90),
 (9,108,130),
 (10,103,80),

(2,104,100),
(11,104,90),
(12,104,80),
(13,104,70),
(14,104,100),
(15,104,75),
(3,105,100),
(16,105,120),
(17,105,100),
(18,105,90),
(19,105,80),
(20,105,70),
(4,106,100),
(21,106,120),
(22,106,100),
(23,106,90),
(24,106,80),
(25,107,100),
(5,107,100),
(26,107,110),
(27,107,90),
(28,107,80),
(29,107,100),
(30,107,90);

```
SELECT e.emp_name, e.salary, d.dept_name
FROM Employee e
INNER JOIN Department d
ON e.dept_id = d.dept_id;
SELECT d.dept_name, e.emp_name
FROM Department d
LEFT JOIN Employee e
ON d.dept_id = e.dept_id;
```

```
SELECT
    e.emp_name AS Employee,
    m.emp_name AS Manager
FROM Employee e
LEFT JOIN Employee m
ON e.manager_id = m.emp_id;
```

```
SELECT
    e.emp_name,
    d.dept_name,
    p.project_name
FROM Employee e
JOIN Department d
ON e.dept_id = d.dept_id
JOIN Employee_Project ep
ON e.emp_id = ep.emp_id
JOIN Project p
ON ep.project_id = p.project_id;
```

```
SELECT
    e.emp_name,
    e.salary,
    e.dept_id
FROM Employee e
WHERE e.salary >
(
    SELECT AVG(e2.salary)
    FROM Employee e2
    WHERE e2.dept_id = e.dept_id
);
SELECT d.dept_name
FROM Department d
```

WHERE EXISTS

```
(  
    SELECT 1  
    FROM Employee e  
    WHERE e.dept_id = d.dept_id  
    AND e.salary > 80000  
);
```

```
SELECT e.emp_id, e.emp_name  
FROM Employee e  
JOIN Employee_Project ep  
ON e.emp_id = ep.emp_id  
WHERE e.dept_id = 1  
AND ep.project_id = 101;
```

```
SELECT e.emp_id, e.emp_name  
FROM Employee e  
WHERE e.dept_id = 1  
AND NOT EXISTS
```

```
(  
    SELECT 1  
    FROM Employee_Project ep  
    WHERE ep.emp_id = e.emp_id  
    AND ep.project_id = 101  
);
```

```
SELECT e.emp_name, d.dept_name  
FROM Employee e  
JOIN Department d  
ON e.dept_id = d.dept_id  
WHERE e.salary > 70000;
```

Output:

ri = K

VO

AI Agent

148 ms

SDNN

Input for the program (Optional)

Output:

emp_name	salary	dept_name
Amit	90000.00	IT
Ravi	60000.00	IT
Sinran	65000.00	IT
Karan	70000.00	IT
Pooja	75000.00	IT
Vikas	58000.00	IT
Neha	85000.00	HR
Anjali	55000.00	HR
Rohit	50000.00	HR
Kavya	52000.00	HR
Manish	60000.00	HR
Sneha	58000.00	HR
Rahul	95000.00	Finance
Deepak	70000.00	Finance
Nisha	62000.00	Finance
Varun	68000.00	Finance
Meena	60000.00	Finance
Sahil	57000.00	Finance
Priya	88000.00	Marketing
Tanya	55000.00	Marketing
Mohit	60000.00	Marketing
Isha	65000.00	Marketing
Akash	58000.00	Marketing
Riya	62000.00	Marketing
Arjun	92000.00	Sales
Naveen	55000.00	Sales
Komal	57000.00	Sales
Yash	68000.00	Sales
Pankaj	65000.00	Sales
Divya	58000.00	Sales

dept_name	emp_name
IT	Karan
IT	Pooja
IT	Vikas
HR	Neha
HR	Anjali
HR	Rohit
HR	Kavya
HR	Manish
HR	Sneha
Finance	Rahul
Finance	Deepak
Finance	Nisha
Finance	Varun
Finance	Meena
Finance	Sahil
Marketing	Priya
Marketing	Tanya
Marketing	Mohit
Marketing	Isha
Marketing	Akash
Marketing	Riya
Sales	Arjun
Sales	Naveen
Sales	Komal
Sales	Yash
Sales	Pankaj
Sales	Divya

Employee	Manager
Amit	NULL
Neha	NULL
Rahul	NULL
Priya	NULL
Arjun	NULL
Ravi	Amit

Light 13

ri = K

VO

AI Agent

148 ms

SDNN

Input for the program (Optional)

Output:

Light 13

Now

History

Save

IT + K

VO

AI Agent

STDM

Input for the program (Optional)

148 ms

DABEER KHAN

Output:

Manish	HR	HR Automation
Sneha	HR	HR Automation
Rahul	Finance	Financial Analytics
Deepak	Finance	Financial Analytics
Rishu	Finance	Financial Analytics
Varun	Finance	Financial Analytics
Meena	Finance	Financial Analytics
Sahil	Finance	Financial Analytics
Priya	Marketing	Marketing Campaign
Hohit	Marketing	Marketing Campaign
Isha	Marketing	Marketing Campaign
Aakash	Marketing	Marketing Campaign
Arjun	Sales	CRM Development
Riya	Marketing	CRM Development
Naveen	Sales	CRM Development
Komal	Sales	CRM Development
Vaish	Sales	CRM Development
Pankaj	Sales	CRM Development
Diloya	Sales	CRM Development
Pooja	IT	Cyber Security

emp_name	salary	dept_id
Amit	90000.00	1
Neha	85000.00	2
Rahul	95000.00	3
Priya	80000.00	4
Arjun	92000.00	5
Karan	70000.00	1
Pooja	75000.00	1
Deepak	70000.00	3
Isha	65000.00	4
Pankaj	65000.00	5

Light 13

Now

History

Save

IT + K

VO

AI Agent

STDM

Input for the program (Optional)

148 ms

DABEER KHAN

Output:

Deepak	70000.00	3
Isha	65000.00	4
Pankaj	65000.00	5

dept_name
IT
HR
Finance
Marketing
Sales

emp_id	emp_name
1	Amit
6	Ravi
8	Karan

emp_id	emp_name
7	Sheran
9	Pooja
10	Vikas

emp_name	dept_name
Amit	IT
Neha	HR
Rahul	Finance
Priya	Marketing
Arjun	Sales
Pooja	IT

Light 13

Now

History

Save